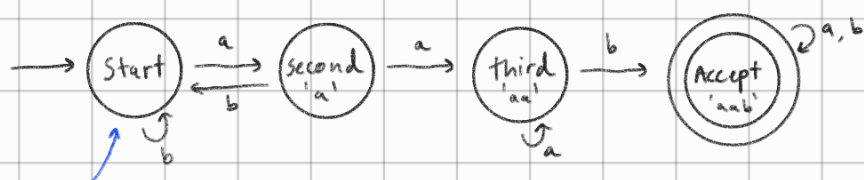


State Diagrams

• aka DFA/Finite State Machine (FSM)

• special directed graph where nodes represent states and edges represent actions

ex. State diagram reads strings of a's and b's, accepts strings that contain "aab" as substring
 goes to accept state



all state diagrams need start state

When "baab" goes thru FSM, should end up in accept state

*labelling nodes meaningfully isn't strictly necessary but is helpful

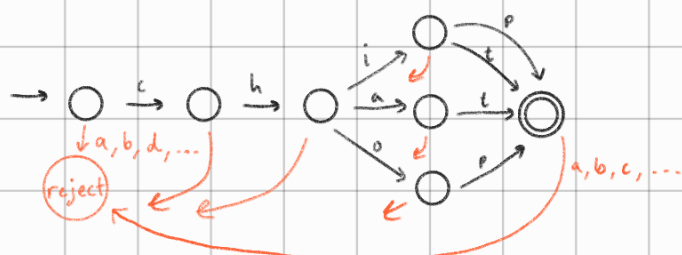
• Accept state means successfully reached condition, has double outline

• If input sequence doesn't reach accept state, input is rejected

ex. Phone lattice, accepts only strings chip, chit, chat, chop. Assume all other transitions go to a dump state

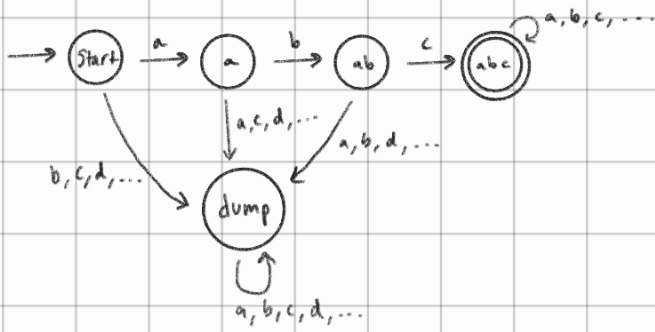


same as:



* don't need this bc we're assuming all other transitions go to dump state

ex. Draw DFA that accepts all strings starting with abc. Strings can be made up of all 26 letters



• Definition: An FSM is made up of the following parts:

- Set of states S including start and terminal(s)/accept(s)/end(s)
one or multiple
- Set of actions A (eg. 26 letters of alphabet)
- Transition function $\delta: S \times A \rightarrow P(S)$

$\delta(\text{state}, \text{action}) = \text{set of states}$

If at start state and read in a , where do I go? $\rightarrow$ set of states output by function δ . CS 173 uses deterministic state machines, so function signature is actually $\delta: S \times A \rightarrow S$ bc we output only one state

non-deterministic state machine: $\rightarrow 0 \xrightarrow{a} 0$
(not included in CS 173)

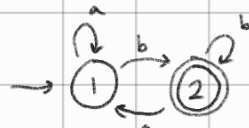
deterministic: $\rightarrow 0 \xrightarrow{a} 0$

ex. States: $\{1, 2\}$ where 1 start, 2 accept

Actions: $\{a, b\}$

Transitions: $\delta: S \times A \rightarrow S$

definition of function δ

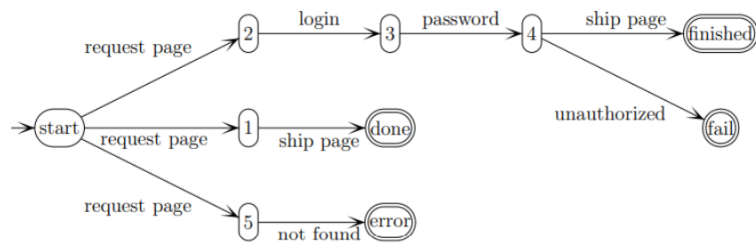
$$\left\{ \begin{array}{l} \delta(1, a) = 1 \\ \delta(1, b) = 2 \\ \delta(2, a) = 1 \\ \delta(2, b) = 2 \end{array} \right\} =$$


accepts strings ending with b

need transition for each state-action pair for a complete function

18.2 A Simple State Diagram

This state diagram represents how a user might retrieve a web page via http.



- Pick two states and write out the directed walk between them. Remember that the full version of the walk contains both a sequence of states and a sequence of actions.
- Let S denote the set of states and A denote the set of actions in the diagram. Explicitly list the contents of S and A .
- Recall that the transition function δ for a state diagram is a function that takes a state/action pair as input and returns the *set* of potential new states. What is the value of δ on the following inputs: $(1, \text{request page})$, $(3, \text{password})$, and $(\text{start}, \text{request page})$?
- How large is $S \times A$? For how many of these values does δ return a value other than the empty set?
- In practice, this state diagram may be more complicated than it needs to be. Can you think of ways to modify the diagram to make it smaller? (For instance, we might decide that we can combine “done” and “finished” into a single state.) Note that modifications to the diagram may alter the transition function, however, you should avoid modifications which change the set of possible sequences of actions.

a) start $\xrightarrow{\text{request page}}$ 2 $\xrightarrow{\text{login}}$ 3

b) $S: \{\text{start}, 1, 2, 3, 4, 5, \text{finished}, \text{fail}, \text{done}, \text{error}\}$

$A: \{\text{request page}, \text{ship page}, \text{not found}, \text{login}, \text{password}, \text{unauthorized}\}$

c) $\delta(1, \text{request page}) = \emptyset$

$\delta(3, \text{password}) = \{4\}$

$\delta(\text{start}, \text{request page}) = \{2, 1, 5\}$

d) $S \times A = 10 \times 6 = 60$

7 values of δ

e) combine 1, 2, and 5